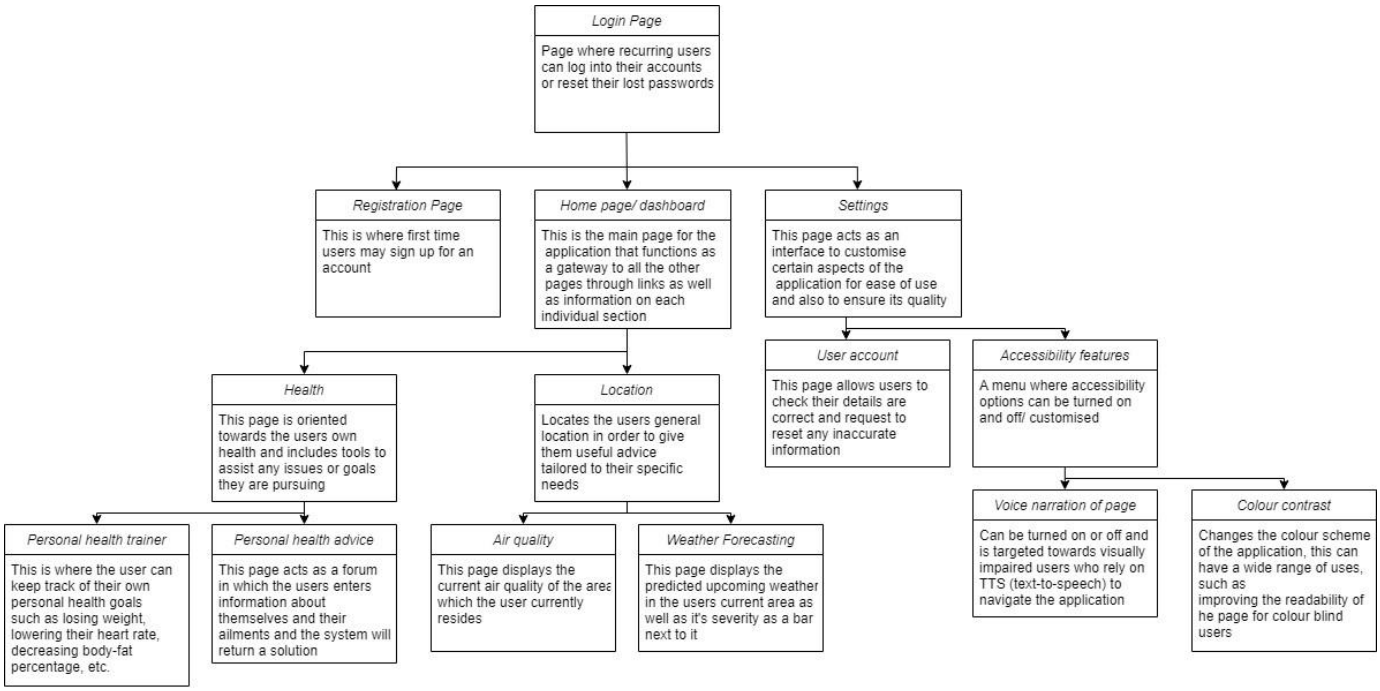
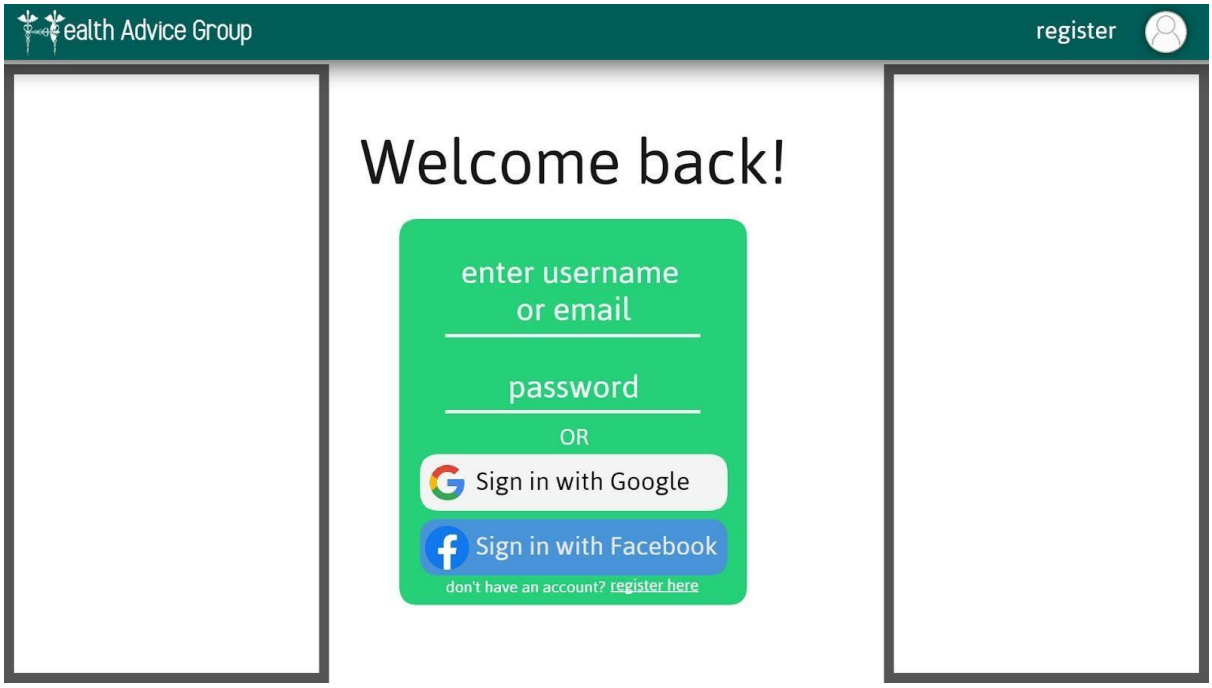
Activity B

Visual/ interface designs:

Website navigation map:



Login page

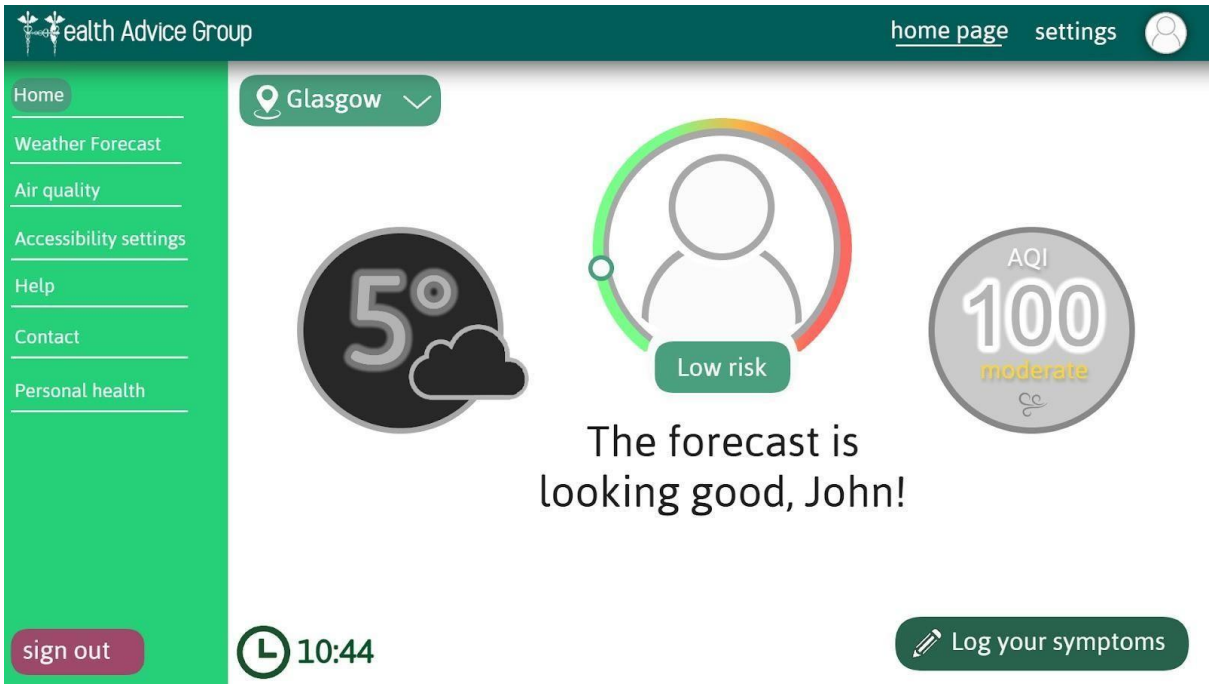


When accessing the website for the first time this is the page that users will be greeted with. In order to avoid confusing new users when entering the site I've gone for a minimalistic approach that will make the website easy to use. Users can sign in using the fields in the centre, they'll be able to sign into an existing account if they have one, or if they haven't made an account they'll be able to access a nearly identical page to register for one. Alternatively the website provides two clear buttons that the user can use to sign in with either Google or Facebook, this can speed up the login process and adds polish to the service. The grey boxes on the side represent where relevant images about the solution will go as they prevent large areas of empty space and can provide further insight into what the website provides, thus aiding the users understanding as a whole.

Page summary:

- Minimalistic and easy to use design
- Company logo for clear branding
- Provides images to add context for the user
- Multiple sign-in options for added accessibility
- Use of Asap font for a sleek modern look

Home page/ dashboard



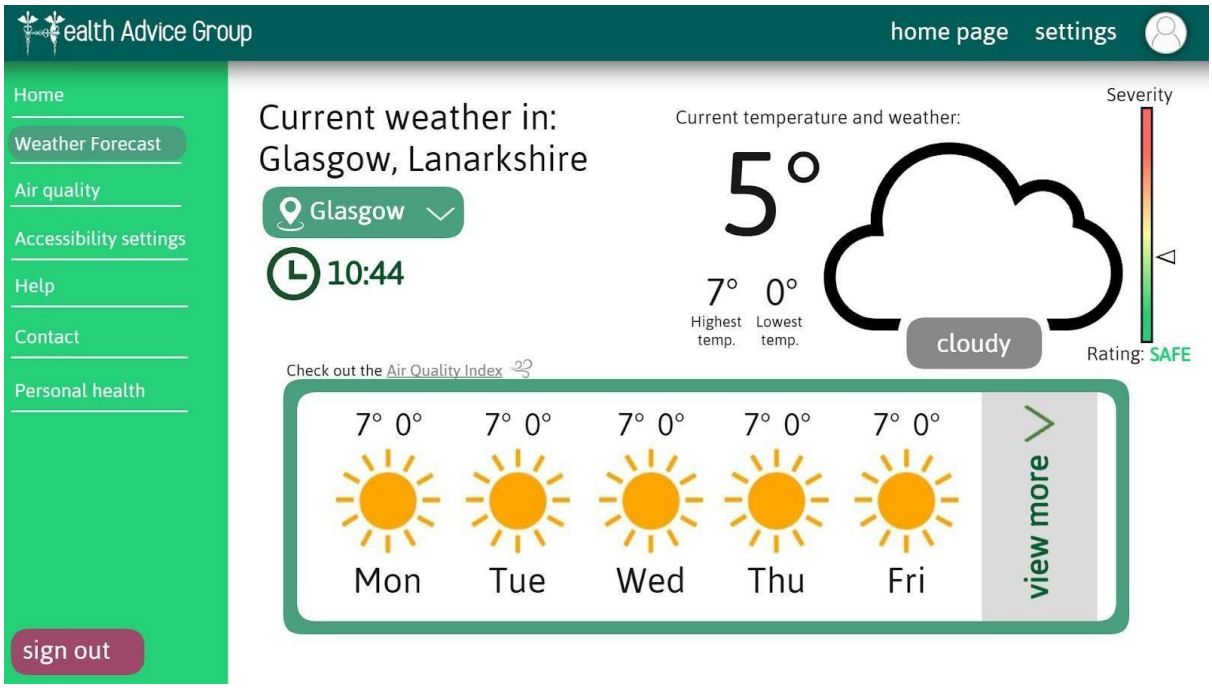
This is the home page/ dashboard that the user will be directed to after they log in, located at the top there is a navbar, containing the company name in the top left corner and the users profile in the top right alongside links to help with navigation, this has been designed this way as many other websites incorporate this same premise and I feel doing this will make the website feel more intuitive to what the users are already used to and familiar with. This navbar will be accessible on every page as it allows the user to navigate throughout the site with ease. Now for the dashboard itself, in order to provide a user interface that, at-a-glance, can inform the user of the air quality among other things, located in the centre of the screen is an indicator of the users overall health, the text underneath will change depending on the risk level and incorporate the users name at the end in order to make the data feel personal to the user. They will also be able to change the location at the top of the screen so the user can see the weather forecasting/ air quality of other places, this can be used for example if the user is going on holiday and wants to make sure it's sunny. On the left side of the screen there is also a dashboard that will also be present on every page like the navbar for ease of use, this acts as a portal to all the other pages of the website. In addition, located either side of the health indicator are a quick summary of the weather and air quality index (AQI) this is so users can access the website quickly and still get the basic information they need without having to go onto another page, however if the user wants more information they may click on the icons and travel to their corresponding page.

Page summary:

- Minimalistic and easy to use design
- Company logo for clear branding

- Uses visual imagery to convey meaning to the user
- Provides clear and simple buttons to navigate the page
- Use of Asap font for a sleek modern look

Weather forecasting



This page presents current and future weather predictions with ease of understanding for the user. The current weather is displayed on the right-hand side with an accompanying icon as well as the current temperature alongside the highest and lowest temperature predicted for that day. An indicator next to it assesses the severity of the day's weather based on temperature and weather type, returning a rating that informs the user of the safety of going outside. Users can change their location to view other weather reports by clicking on the button in the top right, and a panel below displays the upcoming weather for the rest of the week. I have balanced the page layout to eliminate large areas of white space while avoiding a cramped feel.

Page summary:

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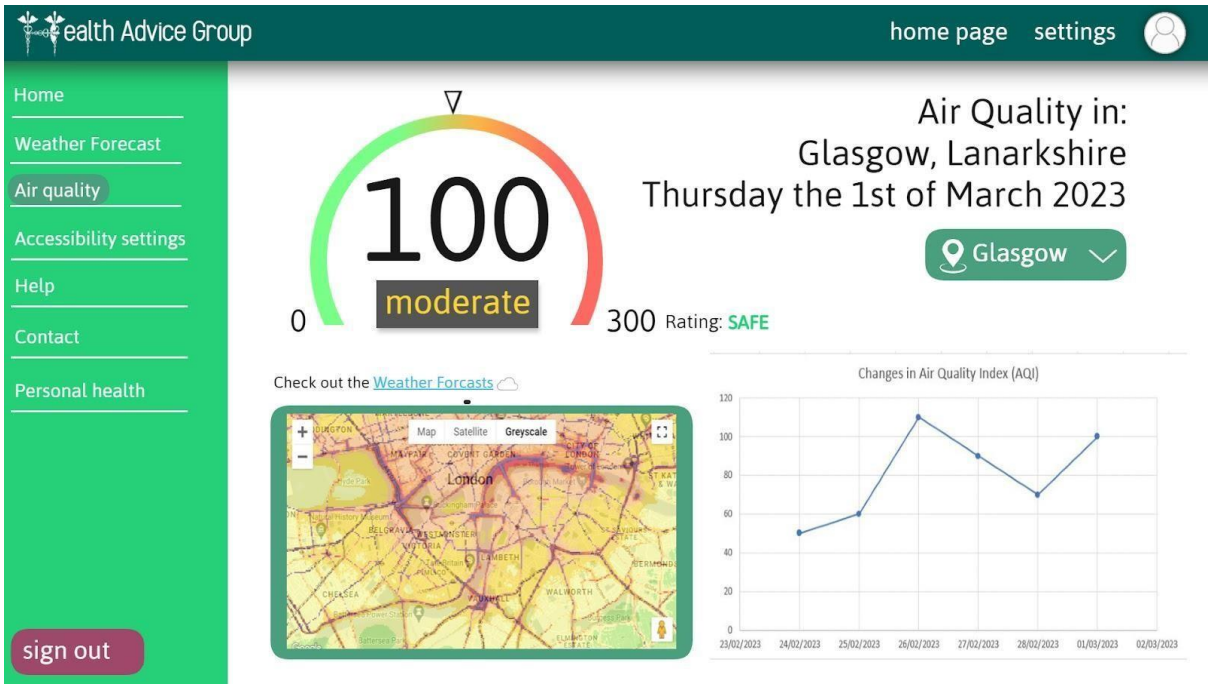
Air quality



This page's role is to provide further insight into the air quality dashboard, allowing the user to access a map to see the air quality of surrounding areas as well as providing a rating of the quality of the air through a combination of graphics and text. In addition the previous average score of the AQI is stored on a graph in the bottom-right that allow the user to see if there are any patterns with the air quality, allowing them to predict the future using reliable information that updates live.

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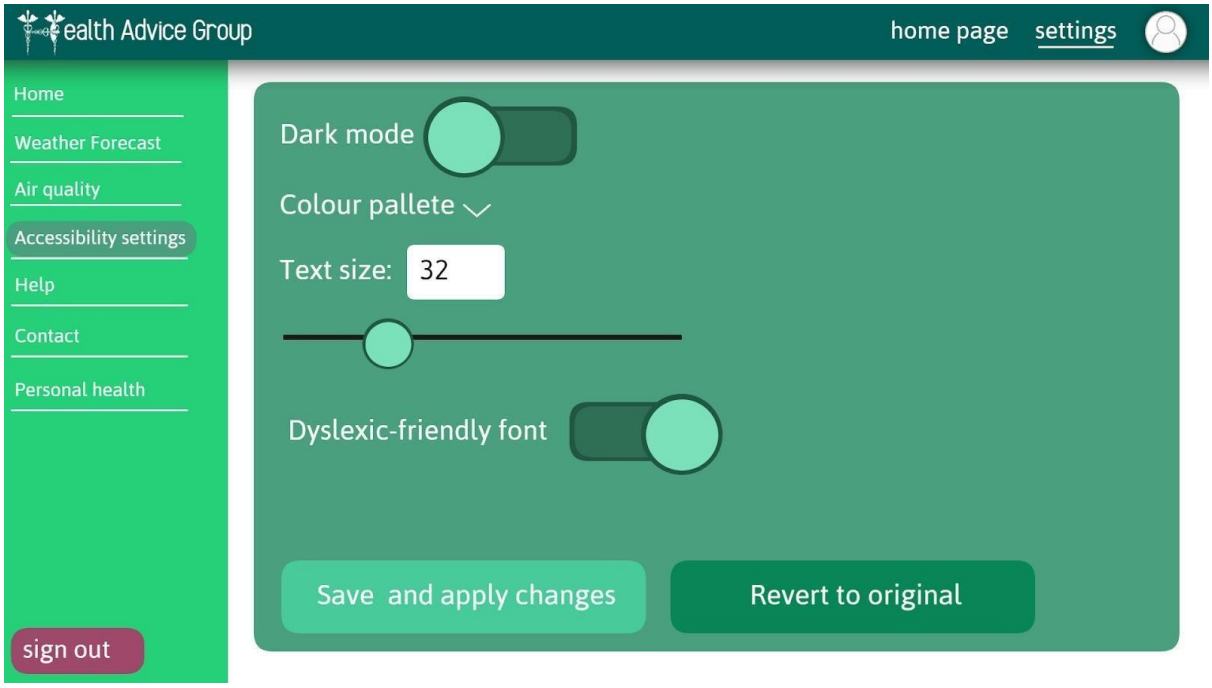
Accessibility settings



This is the settings page, it's here that users will be able to tweak the website in order to suit their specific needs, for example they'll be able to increase the size of the text if they have difficulty reading etc. In the actual solution there'll be more options this is just to provide the basic design layout for the page. Also if at any point the user decides they don't like the changes made they'll be able to revert back to the original through the use of the easy-to-find button in the bottom right corner.

Page summary:

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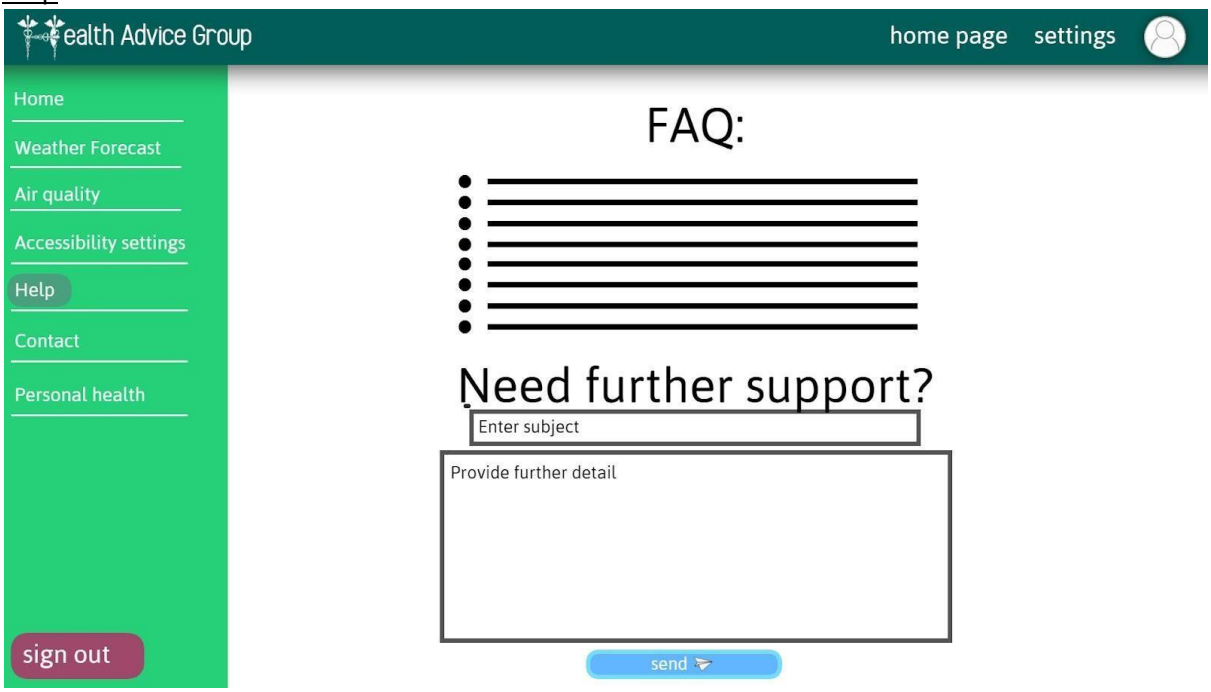
Help



This is the page a user can go to if they have any questions that cannot be found on the main site and need help. To do this they can either send an email to Health Advice Group detailing their issue or they can read the FAQ provided above that answers the most frequently asked questions.

Page summary:

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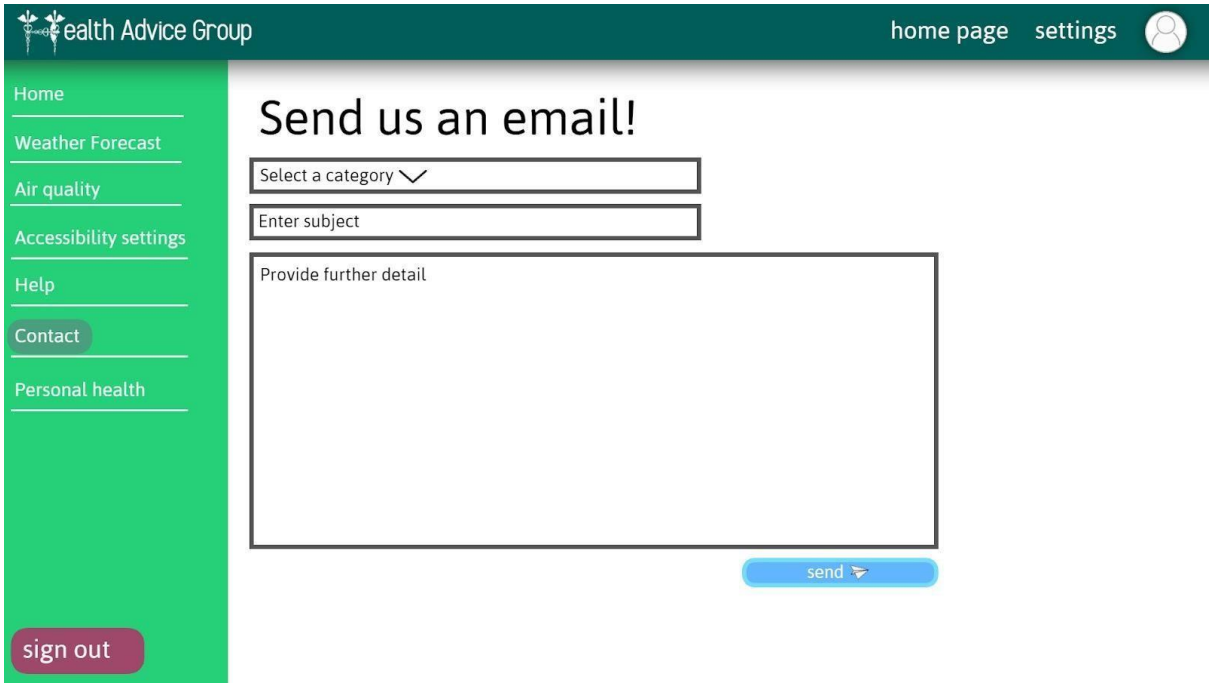
Contact



This page is very similar to the help page however the messages being sent to Health Advice Group do not have to be limited to just questions that the user has, they can select the category of the message, provided by the drop-down box at the top, and then send their message. An example of how this could be used is if someone wanted to reach out to Health Advice Group for a potential partnership, etc.

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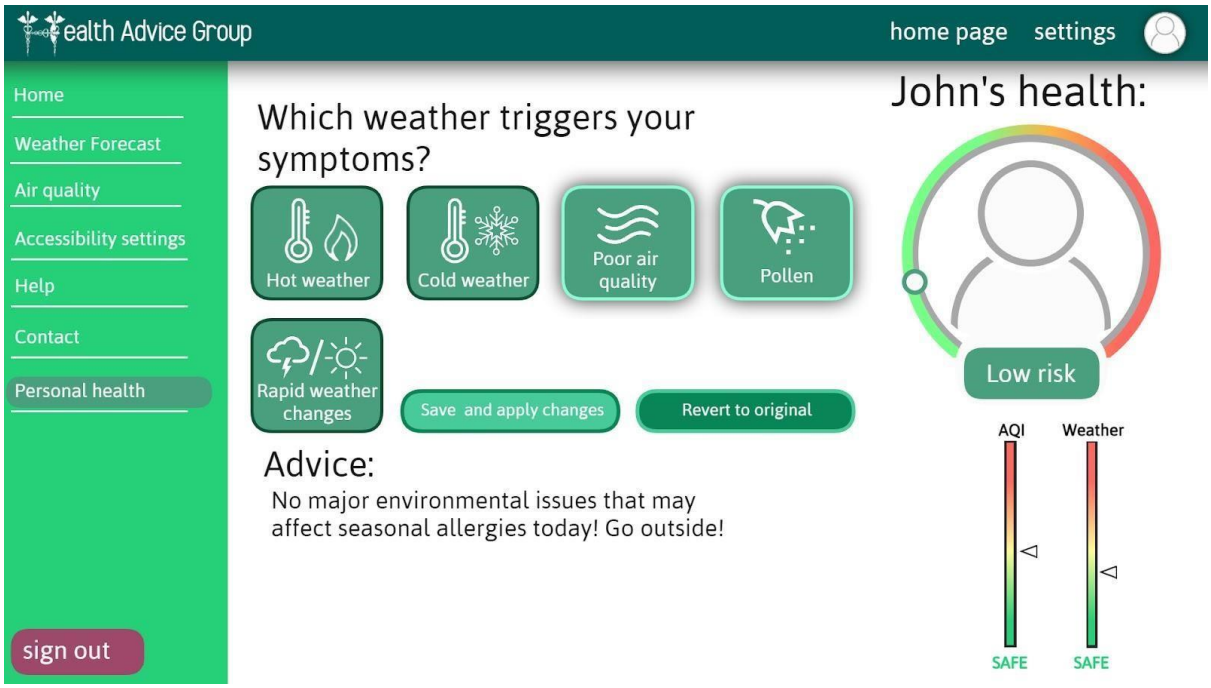
Personal health



This website takes in data from multiple APIs as well as the users input in order to determine how vulnerable the user is to environmental and seasonal issues at any one time, the user may select which triggers may cause their symptoms and look at the indicators on the right-hand side for useful feedback, if that isn't enough then the text at the bottom of the screen will be created automatically by the website in response to the users situation and advise them on what to do next, in the image there are no issues present but if for example the temperature was deemed too low the user would be advised to stay inside/ wear warm clothing, etc.

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Data requirements:

Login	
Field	**Data typ**
userID	int()
username	varChar
password	varChar
email	varChar

HealthInfo	
Field	**Data type**
userID	int()
hotWeather	boolean
coldWeather	boolean
poorAirQuality	boolean
pollen	boolean
rapidWeather	boolean
advice	varchar(100)
temperature	int()
weatherType	varChar(10)
airQuality	Int(3)

Variable name	Function	Data type	Justification
$username	Assigns the users input to a variable for the username	String	Allows any character to be entered instead of limiting it to just numbers like an integer
$usernamePresent	Checks if the ‘username’ field is filled in	String	Prevents an empty input
$usernameError	Outputs an error message is username is incorrect	String	Informs the user what went wrong so they can fix it

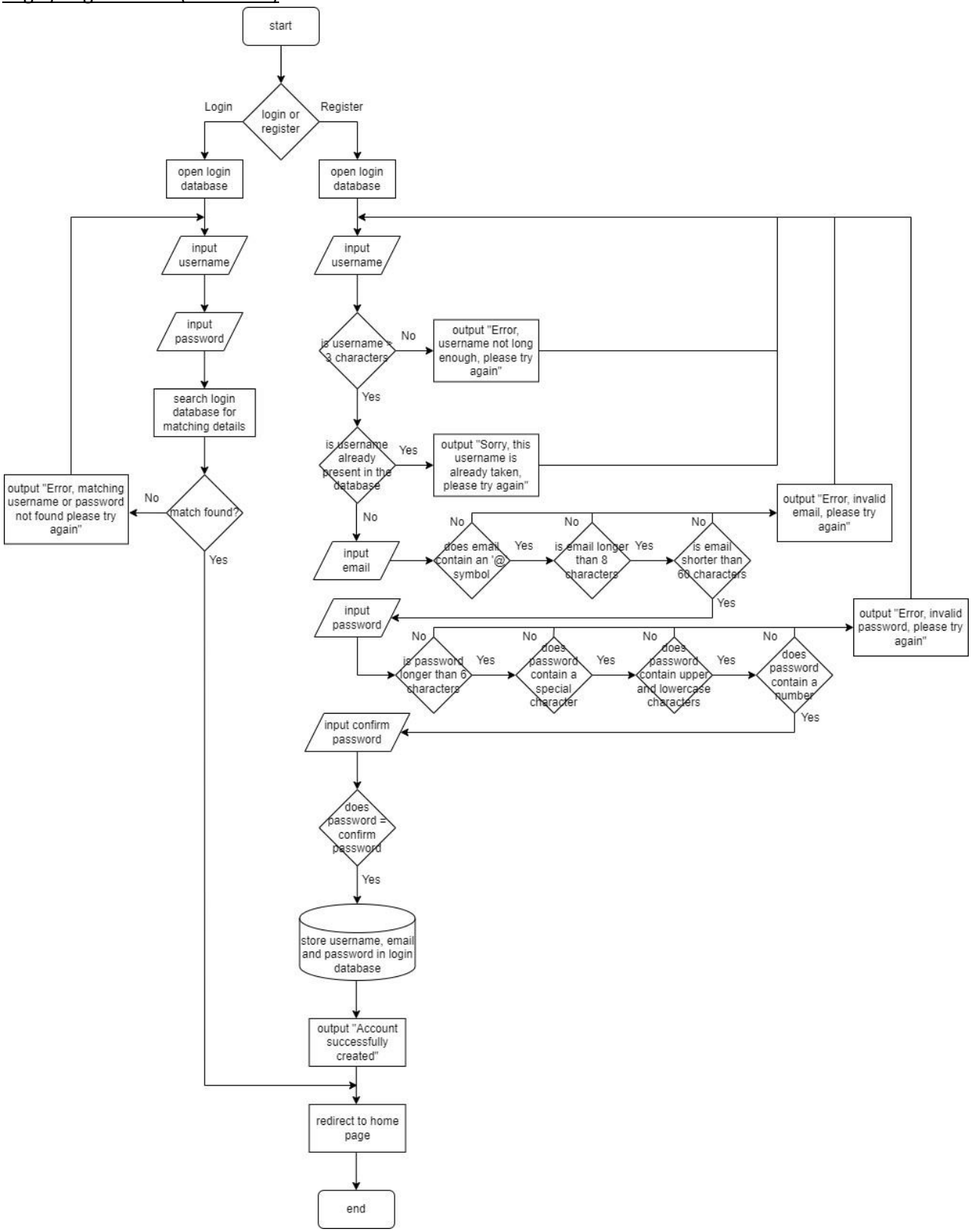
$password	Assigns the users input to a variable for the password	String	Allows any character to be entered instead of limiting it to just numbers like an integer
$passwordPresent	Checks if the ‘password’ field is filled in	String	Prevents an empty input
$passwordConfirm	Assigns the users input to a variable that should match the password	String	Ensures that the user is able to replicate the same password so they don’t face trouble when signing into the site in the future
$passwordPresentConfirm	Checks if the ‘confirm password’ field is filled in	String	Prevents an empty input
$passwordError	Outputs an error message if password or passwordConfirm is wrong	String	Informs the user what went wrong so they can fix it
$email	Assigns the users input to a variable for the email address	String	Having the users email address allows for extra security features like 2factor authentication
$emailValid	To determine if the email entered by the user is valid by using other variables as checks	Boolean	As either the validity of the email is TRUE or FALSE
$emailSymbolCheck	Validate the email contains an ‘@’ symbol and has it in the correct position	String	If the email does not include an @ symbol then there can be no accompanying mail server meaning the email is invalid
$emailLengthCheck	Ensures the entered emails length is valid, meaning it is longer than 8 characters and shorter than 60	String	To help validate the length of the email

$userID	Assigns a unique number to each individual user	Integer	Prevents the system from getting confused about which data belongs to which user
$hotWeather	Stores the users input to this variable	Boolean	Allows the user to tell the website whether or not they are especially susceptible to hot weather
$coldWeather	Stores the users input to this variable	Boolean	Allows the user to tell the website whether or not they are especially susceptible to cold weather
$poorAirQuality	Stores the users input to this variable	Boolean	Allows the user to tell the website whether or not they are especially
			susceptible to poor air quality
$pollen	Stores the users input to this variable	Boolean	Allows the user to tell the website whether or not they have hayfever
$rapidWeather	Stores the users input to this variable		Allows the user to tell the website whether or not they are especially susceptible to rapid changes in the weather
$advice	Displays blocks of text depending on how at risk the user is determined to be	String	Takes into account the users ailments and provides advice based on it
$temperature	Assign the current temperature to a variable	Integer	Informs the user of the current temperature
$weatherType	Assign the current weather type to a variable	String	Informs the user of the current type of weather
$airQuality	Assign the current air quality to a variable	Integer	Informs the user of the air quality

A selection of algorithm designs:

Occupational Specialism Task 1 – Design documents

Login/ registration: (flowchart)



This diagram represents the process the user will have to undergo when creating a new account or signing into an already pre-existing one where in both cases they will communicate with a database to transfer data or validate it.

Occupational Specialism Task 1 – Design documents

Login/ registration: (pseudocode)

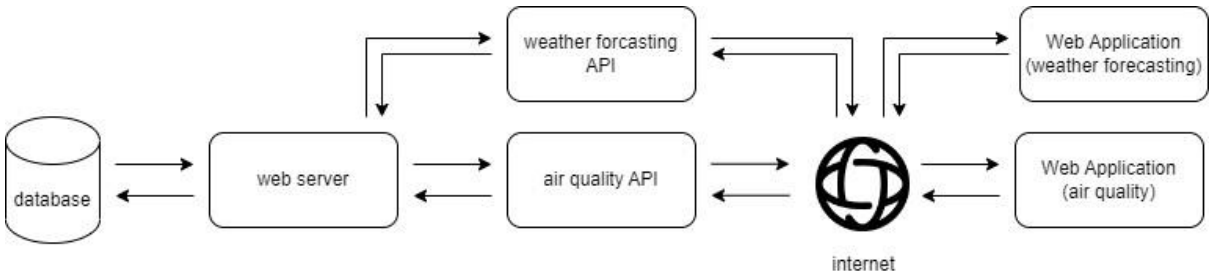
```
START
Function login():
Open login database
Input username
Input password
SQL:
SELECT * FROM users WHERE username = '{username}' AND password = '{password}'; If
 username OR password != login database:
        Print("Error, matching username or password not found, please try again")
Else:
        Print("Login successful!")

Function registration():
Open login database Successful = 0
special_character =
["*","&","£","!","?","/","\"]
WHILE successful == 0:
        Input username
        Input password
        Input email
        If username.length > 3:
                If password.length > 6 && password contain(special_character, integer) &&
        password.Upper != TRUE:
                        If email.contain("@") && email.length > 8 && email.length < 60:
                                STORE username, password AND email in login database;
                                Print("Account successfully created")
                                Successful = 1;
                        Else:
                                Print("Error, invalid email, please try again")
                Else:
```

```
                    Print("Error, invalid password, please try again")
        Else:
                Print("Error, username not long enough, please try again")
END
```

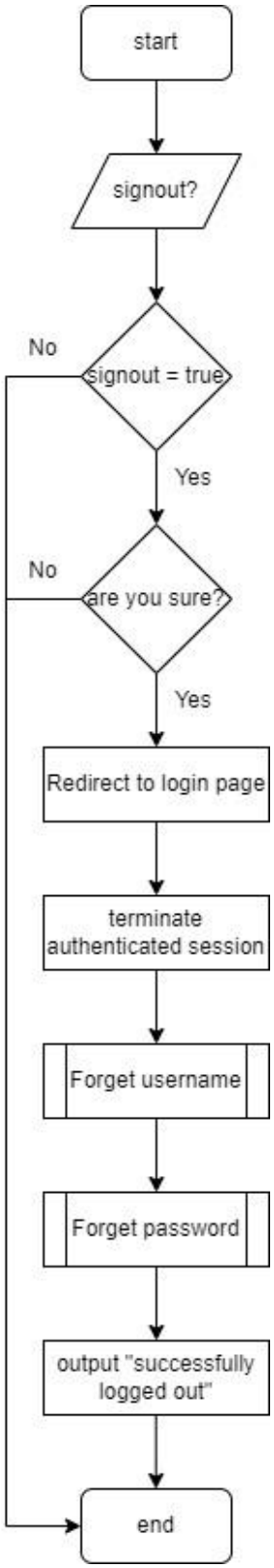
This pseudocode is a version of the diagram above, helping the developers to visualise how the login system will be implemented using two programming languages, python for the skeleton of the program and SQL to communicate with the databases.

How APIs will exchange data with the website:



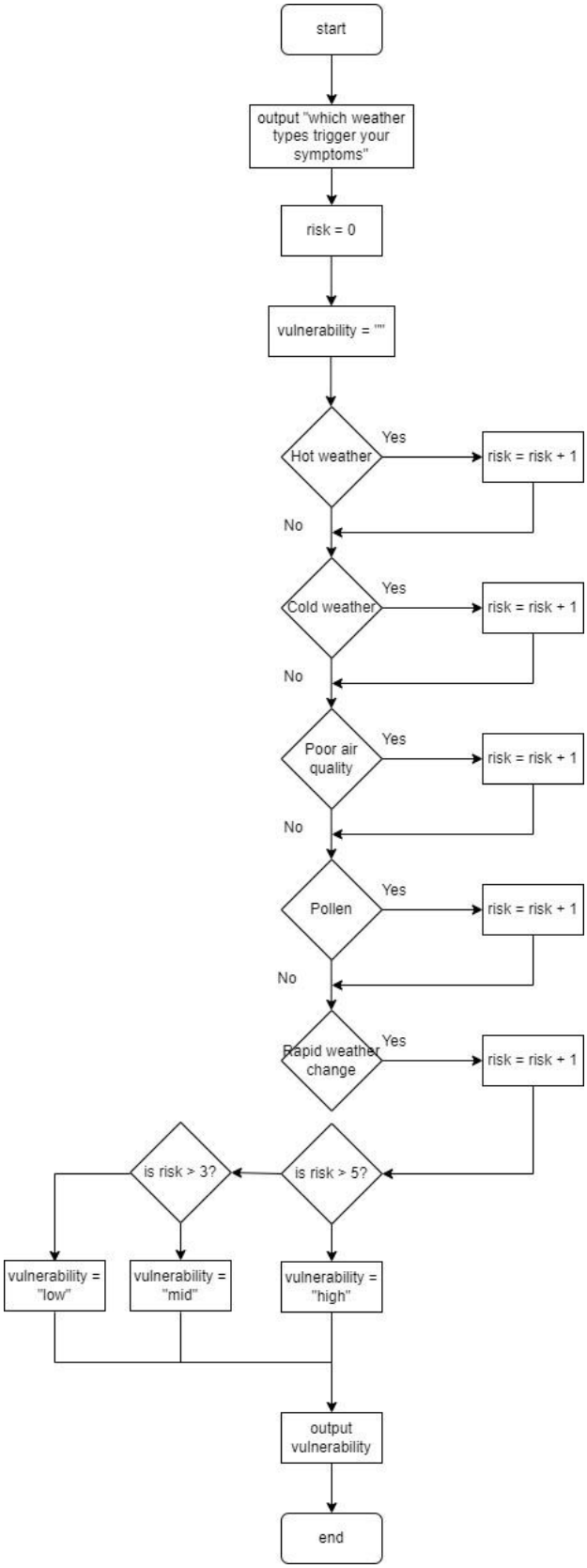
This diagram shows the process APIs will take in order to get reliable and up-to-date information delivered to the website

Signing out:



This diagrams purpose is to visualise the process that occurs when a user signs out of their account

<u>Weather vulnerability check: (Flowchart)</u>



This diagrams calculates the vulnerability of the user by asking them about their vulnerabilities and incrementing the vulnerability variable by 1 each time a symptom is detected.

Test strategy:

Testing method	White box testing

Purpose	Used to ensure the backend/ infrastructure of the website functions in the intended way
Who will carry out the tests	Programmer (me)
Testing data	All inputs and outputs are recorded and tested with every single type of data, these include: erroneous data, null data, normal data, etc.
Test criteria	Checks the actual output of the tests against the intended output, if the result is as expected the test is passed
When will testing be carried out	Whenever a new section of the website is completed the inputs and outputs will be accounted for and then they may be tested with every data type to ensure they function in the intended way
Test data outcome	The outcome of the tests will be compared against what the expected outcome was and if they do not match the test has failed and the necessary changes/ improvements to the website will be made.

Testing method	Black box testing
Purpose	Tests all the inputs and outputs of the website to ensure that they function within the intended way
Who will carry out the tests	Developer (me)
Testing data set	All inputs and outputs will be recorded
Test criteria	Checks the actual output of the tests against the intended output, if the result is as expected the test is passed
When will testing be carried out	Whenever a new section of the website is completed the inputs and outputs will be accounted for and then they may be tested to ensure they function in the intended way
Time testing will take	Each test will take less than half an hour, however this may vary depending on the number of input and outputs that are part of each test
Test data outcome	Once testing is complete the actual results of the tests will be compared to what the intended results were. Based on this comparison changes/ improvements can be made to the website accordingly.

Date of test	Component to be tested	Type of test to be carried out	Prerequisites and dependencies
-	When signing up entering an invalid password will result in an error	Black box testing	Access to the website Computer Incorrect password
-	When signing up entering an invalid username will result in an error	Black box testing	Access to the website Computer Incorrect username
-	When signing up entering if the confirmed password field does not match the password it will result in an error	Black box testing	Access to the website Computer Incorrect confirm password input
-	When signing up entering an invalid email will result in an error	Black box testing	Access to the website Computer Incorrect email
-	When logging in if the username and password are correct the user will be redirected to the home page	White box testing	Access to the website Computer Correct username: 'Test123' Correct password 'admin'
-	When a button linking to a different page in the dashboard is clicked the user will be redirected to that page	Black box testing	Access to the website Computer Clicking a button that links to a different page in the dashboard
-	When the user clicks "sign out" they'll be redirected to the login page and have to sign in again to access the main website	White box testing	Access to the website Computer Already signed in on an account Clicking on the "sign out" button
-	When a user clicks on the weather widget on the home page they'll be redirected to the weather forecast page for more information	White box testing	Access to the website Computer Already signed in on an account Clicking on the weather widget

-	When a user clicks on the air quality widget on the home page they'll be redirected to the air quality page for more information	White box testing	Access to the website Computer Already signed in on an account Clicking on the air quality widget
-	When a user clicks on the overall health widget on the home page they'll be redirected to the	White box testing	Access to the website Computer Already signed in on an account Clicking on the overall health widget
	personal health page for more information		
-	When a user creates their account they'll receive an email confirming it was them	White box testing	Access to the internet Access to the website Computer Creating an account
-	The user will be able to input their ailments on the personal health page and it will have an effect on the overall health risk for the user	Black box testing	Access to the website Computer Clicking the buttons on the personal health page